

Class10: Structural Bioinformatics 1

Yassaman Fakhari (PID A18541145)

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Background

The main repository of high-resolution structural data on biomolecules is called the **Protein Data Bank** (PDB).

PDB Statistics

What is in the PDB in terms of molecule type and structure determination method?

Read a CSV file of current PDB stats obtained from <https://www.rcsd.org/stats/summary>

```
pdb <- read.csv("Data Export Summary.csv")
pdb
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	180,758	23,111	12,813	348	229
2	Protein/Oligosaccharide	10,488	3,741	34	8	11
3	Protein/NA	9,205	6,751	287	26	8
4	Nucleic acid (only)	3,154	250	1,578	3	15
5	Other	178	27	35	4	0
6	Oligosaccharide (only)	11	0	6	0	1
	Neutron Other Total					
1	84 32	217,375				
2	1 0	14,283				

3	0	0	16,277
4	3	1	5,004
5	0	0	244
6	0	4	22

Q1 What percentage of structures in the P

```
pdb$X.ray
```

```
[1] "180,758" "10,488" "9,205" "3,154" "178" "11"
```

We could also use a different import function for this CSV that speaks American (i.e. deals with commas in numbers in a comma separated value file).

```
library(readr)

pdb <- read_csv("Data Export Summary.csv")
```

```
Rows: 6 Columns: 9
-- Column specification -----
Delimiter: ","
chr (1): Molecular Type
dbl (4): Integrative, Multiple methods, Neutron, Other
num (4): X-ray, EM, NMR, Total

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
n.tot <- sum(pdb$Total)
```

This print out above ‘pdb\$X.ray’ is “character” not “numeric”. Therefore I can’t do math with it. We need to fix this...

```
# We want to get rid of commas:
x <- pdb$X.ray
```

Warning: Unknown or uninitialised column: ‘X.ray’.

```
tmp <- sub(",", "", x)
sum( as.numeric(tmp))
```

```
[1] 0
```

We could make a function to do this:

```
rm.comma <- function(x) {
  tmp <- sub(",", "", x)
  sum( as.numeric(tmp))
}
```

```
n.xray <- rm.comma(pdb$X.ray)
```

Warning: Unknown or uninitialised column: `X.ray`.

```
n.em <- rm.comma(pdb$EM)
n.tot <- rm.comma(pdb$Total)

n.xray/n.tot*100
```

```
[1] 0
```

```
n.em/n.tot*100
```

```
[1] 13.38046
```

Q How many total protein structures are there in this dataset?

```
pdb$Total[1]
```

```
[1] 217375
```

The total number of protein sequences in UniPort is 202,556,314.

```
217375/202556314*100
```

```
[1] 0.1073158
```

Key-Point: We have a very, very small structural coverage of known proteins (~0.1%). Most structures we know about (~80%) come from one method (X-ray crystallography).

Visualizing PDb Data with Mol-Star

Main stand alone web version with all features is at <https://molstar.org/viewer/>.



Figure 1: The HIV-protease enzyme is a homodimer with two chains

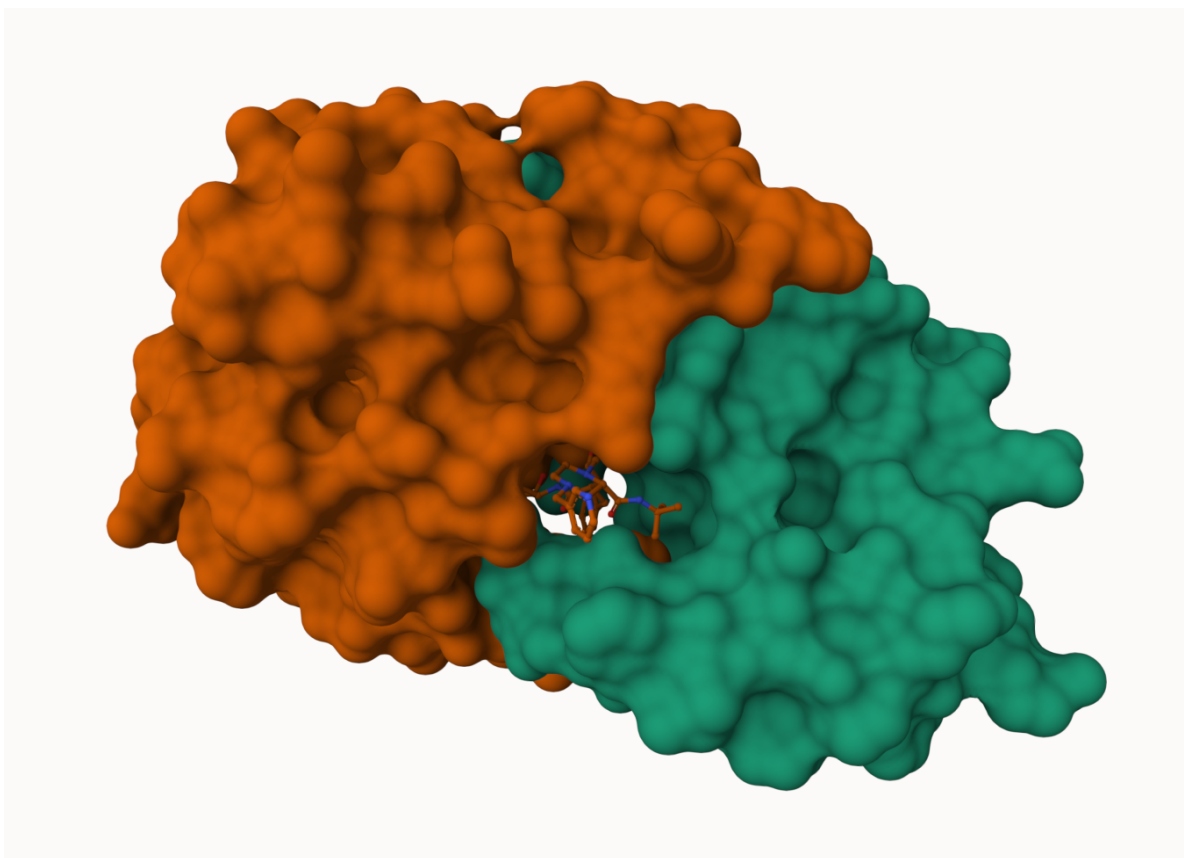


Figure 2: Surface Display showing the Binding Cleft Site for the Inhibitor (Drug) Molecule

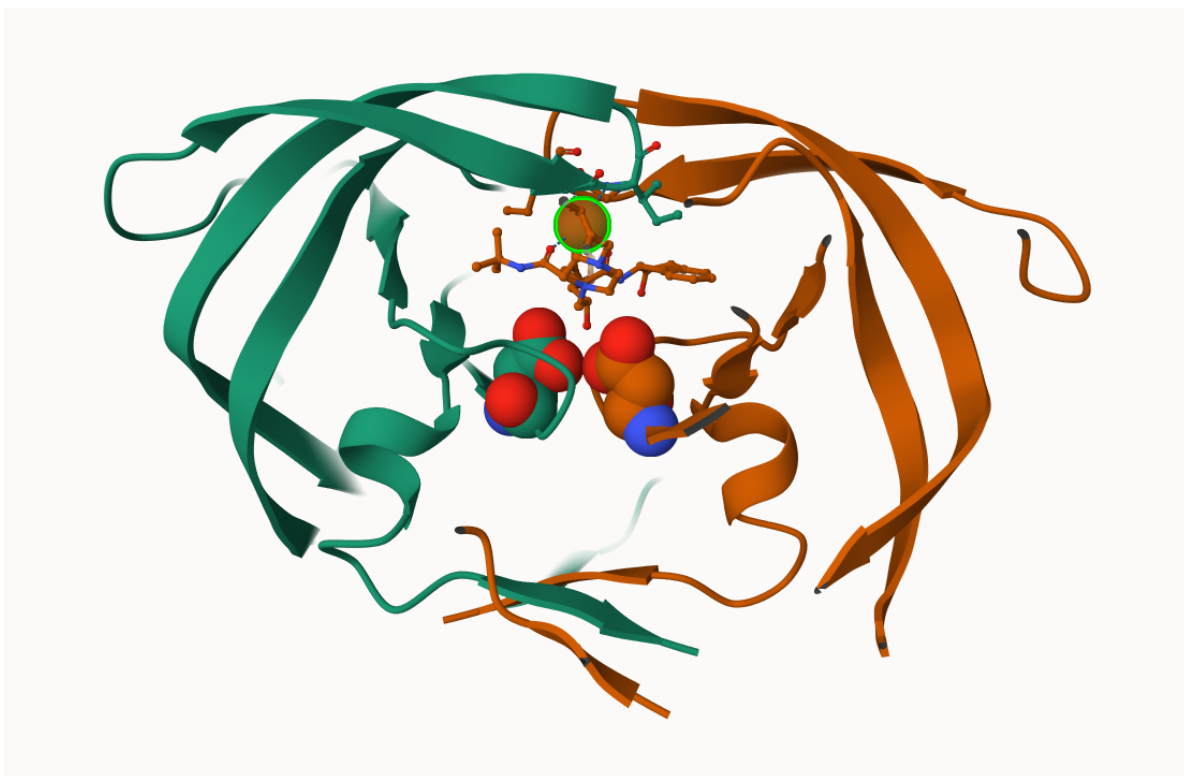


Figure 3: Spacefill Display of Catalytic ASp25 amino acids and key binding site molecule